

Planar dynamical systems

May 15, 2026

Planar dynamical systems (n=2)

We consider differential equations of the form

$$\dot{X} = f(X)$$

$$\begin{aligned} n=1 & \quad \dot{x} = f(t, x) \\ n=2 & \quad \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = A \cdot \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

where $f : \mathbb{R}^{\textcircled{2}} \rightarrow \mathbb{R}^{\textcircled{2}}$ is a given C^1 function.

In the previous lecture it was given the existence and uniqueness theorem that mainly assures that, for any **initial state** $\eta \in \mathbb{R}^2$ there exists a unique solution of (1) with $X(0) = \eta$. This solution is denoted by $t \mapsto \varphi(t, \eta)$ and it describes the motion of (1) that initiates at η . The **orbit** is $\gamma_\eta = \{\varphi(t, \eta) : t \in I_\eta\}$.

The function φ of both variables (t, η) is called the **flow** of (1).

$$\varphi(t, \eta) = (x(t, \eta), y(t, \eta)) : n=2$$

Also, the following notions have been presented:

equilibrium point (constant solution, stationary point, steady state)

orbit (all the states of the system during one motion)

attractor equilibrium point (when starting nearby, the system tend to it in the future)

repeller equilibrium point (the same as "attractor" but reversing the arrow of time)

stable equilibrium point (the system remains close to it after starting nearby)

phase portrait (graphical representation of the orbits)

The case $n = 1$ was discussed.

The main goal of the theory is to represent the phase portrait.

Why? Because from the phase portrait one can read (deduce) the main properties of the solutions of the differential system, in other words, the long-term behavior of the dynamical system.

How? If it is possible, **without** knowing the expressions of the solutions (since, in general, it is impossible to find them).

In the case $n = 1$, we already know, it is possible to represent the phase portrait without solving the differential equation. We need only to study the sign of the function f from $\dot{x} = f(x)$.

The phase portrait can be more complex in higher dimensions and the study is more complicated, of course. Today we consider $n = 2$. First we study the linear systems.

The type of linear planar systems

$$\det(A - \lambda I_2) = 0$$

$$\dot{X} = AX \tag{2}$$

where $A \in \mathcal{M}_2(\mathbb{R})$ with $\det(A) \neq 0$.

Denote by $\lambda_1, \lambda_2 \in \mathbb{C}$ the eigenvalues of A . We know that $\det(A) = \lambda_1 \lambda_2$. Note that, since $\det(A) \neq 0$, the only equilibrium point of (2) is the origin, and, in addition, $\lambda_1 \neq 0$ and $\lambda_2 \neq 0$.

We will classify the linear planar systems using words (node, saddle, center, focus) that reflect the **geometry** of the orbits, but using pure **algebraic** criteria (the nature of the eigenvalues).

The type of linear planar systems

Definition

We say that the linear planar system (2) is a

node when $\lambda_1, \lambda_2 \in \mathbb{R}$ and they have the same sign.

saddle when $\lambda_1, \lambda_2 \in \mathbb{R}$ and they have opposite signs.

center when $\lambda_{1,2} = \pm i\beta$, with $\beta > 0$, $\beta \in \mathbb{R}$.

focus when $\lambda_{1,2} = \alpha \pm i\beta$, with $\alpha \neq 0$, $\beta > 0$, $\alpha, \beta \in \mathbb{R}$.

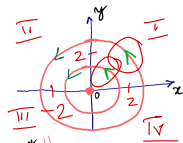
Phase portraits of linear planar systems

We intend to sketch the phase portrait of the following (simple) linear planar systems.

a) center ($\lambda_{1,2} = \pm i$)

$$\begin{matrix} x \downarrow & y \downarrow \\ \dot{x} < 0 & , & \dot{y} < 0 \end{matrix}$$

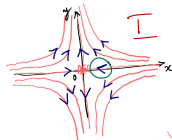
$$\begin{cases} \dot{x} = -y, \\ \dot{y} = x \end{cases}$$



b) saddle ($\lambda_1 = -1 < 0$, $\lambda_2 = 1 > 0$)

$$x = 2 \quad y = 0$$

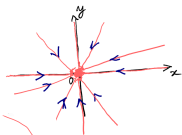
$$\begin{cases} \dot{x} = -x, \\ \dot{y} = y \end{cases}$$



$$\begin{matrix} \dot{x} < 0 & y = 0 \\ x < 0 & \dot{y} < 0 \end{matrix}$$

c) node ($\lambda_1 = -1 < 0$, $\lambda_2 = -1 < 0$)

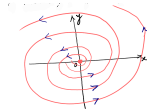
$$\begin{cases} \dot{x} = -x, \\ \dot{y} = -y \end{cases}$$



d) focus ($\lambda_{1,2} = 1 \pm i$)

$$\begin{matrix} L > 0 \\ L < 0 \end{matrix}$$

$$\begin{cases} \dot{x} = x - y, \\ \dot{y} = x + y \end{cases}$$



A node

c) $\dot{x} = -x$, $\dot{y} = -y$. Find the flow. Using the definition of attractor, justify that the origin is a (global) attractor. Note that $\mathbb{R} \times (0, \infty)$ is an invariant set. Find the expression of a first integral in $\mathbb{R} \times (0, \infty)$.

In order to find the flow we have to consider the IVP

$$\text{IVP: } \begin{cases} \dot{x} = -x, & \frac{dx}{dt} = -x \\ \dot{y} = -y, & \frac{dy}{dt} = -y \end{cases} \quad x(0) = \eta_1, \quad y(0) = \eta_2$$

Handwritten notes: $\frac{dx}{dt} = -x$, $\frac{dy}{dt} = -y$, $\ln|x| = -t + C$

for each fixed $\eta = (\eta_1, \eta_2) \in \mathbb{R}^2$. Calculations yields that the flow $\varphi : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ has the expression

$$\varphi(t, \eta_1, \eta_2) = (\eta_1 e^{-t}, \eta_2 e^{-t}).$$

Handwritten notes:
 $x(t) = e^{-t} \cdot C$
 $x(0) = C = \eta_1$
 $x(t, \eta_1) = \eta_1 \cdot e^{-t}$

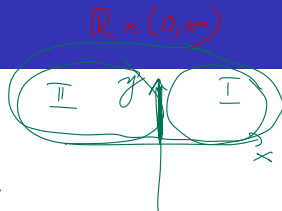
The origin is a global attractor since $\lim_{t \rightarrow \infty} \varphi(t, \eta_1, \eta_2) = (0, 0)$ for all $(\eta_1, \eta_2) \in \mathbb{R}^2$.

A node

$$H: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$H(x, y)$$

$$\varphi(t, \eta_1, \eta_2) = (\eta_1 e^{-t}, \eta_2 e^{-t}).$$



$\mathbb{R} \times (0, \infty)$ is an invariant set since $(\eta_1, \eta_2) \in \mathbb{R} \times (0, \infty) \implies \eta_2 > 0 \implies \eta_2 e^{-t} > 0, \forall t \in \mathbb{R} \implies \gamma_\eta \subset \mathbb{R} \times (0, \infty)$.

In order to find the expression of a first integral in $\mathbb{R} \times (0, \infty)$ we have to recall that the time t must disappear after we replace

$$x = \eta_1 e^{-t} \text{ and } y = \eta_2 e^{-t} \Leftrightarrow \frac{\eta_2 e^{-t}}{\eta_1 e^{-t}} = \frac{\eta_2}{\eta_1} = \text{constant} \Rightarrow H(x, y) = \frac{y}{x} \text{ or } \frac{x}{y}$$

Check that $H: \mathbb{R} \times (0, \infty) \rightarrow \mathbb{R}$, $H(x, y) = \frac{x}{y}$ is a first integral in $\mathbb{R} \times (0, \infty)$. Note that the same expression also defines a first integral in $\mathbb{R} \times (-\infty, 0)$.

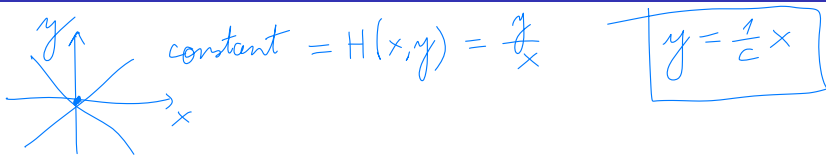
$$H(x, y)$$

Another way to compute the first integral is:

$$\frac{dy}{dx} = -\frac{y}{x} = -\frac{y}{x} = -\frac{y}{x} \Leftrightarrow \frac{dy}{y} = -\frac{dx}{x} \mid \Leftrightarrow \ln \left| \frac{y}{x} \right| = C \Leftrightarrow \frac{y}{x} = C$$

$$\frac{dy}{y} = -\frac{dx}{x} \mid \Leftrightarrow \ln \left| \frac{y}{x} \right| = C \Leftrightarrow \frac{y}{x} = C$$

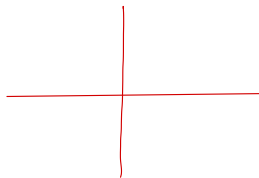
A node



In order to find the shape of the orbits, note that the level curves of H are $y = \frac{1}{c}x$, $c \in \mathbb{R}$. There are the lines that contain the origin.

On each orbit, the arrow must point to the origin since it is a global attractor.

$$\frac{y}{x} = c \quad \text{or} \quad \frac{x}{y} = c \iff y = \frac{1}{c}x$$



A focus

d) $\dot{x} = x - y$, $\dot{y} = x + y$ Transform the system to polar coordinates. Find the flow in polar coordinates. Find the shape of the orbits. Represent the phase portrait.

Recall that, for each $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$ there exists a unique $(\rho, \theta) \in (0, \infty) \times [0, 2\pi)$ such that $x = \rho \cos \theta$, $y = \rho \sin \theta$ (or, equivalently, $x + iy = \rho e^{i\theta}$). Note that we have the reversed relations $\rho^2 = x^2 + y^2$, $\tan \theta = \frac{y}{x}$.

We consider now the unknown functions $(\rho(t), \theta(t))$ such that, for each t , these are the polar coordinates of $(x(t), y(t))$, that is

$\rho^2(t) = x^2(t) + y^2(t)$, $\tan \theta(t) = \frac{y(t)}{x(t)}$. After taking the derivative with respect to t , we get

A focus

$$\rho \dot{\rho} = x\dot{x} + y\dot{y}, \quad \frac{\dot{\theta}}{\cos^2 \theta} = \frac{x\dot{y} - y\dot{x}}{x^2}.$$

$$\cancel{2\rho \cdot \dot{\rho}} = \cancel{2x} \dot{x} + \cancel{2y} \dot{y}$$

Now we replace $\dot{x} = x - y$, $\dot{y} = x + y$. We get

$$\rho \dot{\rho} = x\dot{x} + y\dot{y} = x(x - y) + y(x + y) = \underline{x^2 + y^2 = \rho^2}$$

$$\dot{\rho} = \frac{\rho^2}{\rho}$$

$$\frac{\dot{\theta}}{\cos^2 \theta} = \frac{x\dot{y} - y\dot{x}}{x^2} = \frac{x(x+y) - y(x-y)}{x^2} = \frac{\rho^2}{\rho^2 \cos^2 \theta} = \underline{\frac{1}{\cos^2 \theta}}. \quad \checkmark$$

$$\dot{\theta} = \frac{\cancel{\cos^2 \theta}}{\cancel{\cos^2 \theta}} = 1$$

We arrive to a simple system in polar coordinates

$$\dot{\rho} = \rho, \quad \dot{\theta} = 1.$$

If we impose the initial conditions $\rho(0) = \rho_0$, $\theta(0) = \theta_0$, we immediately see that $\rho(t) = \rho_0 e^t$, $\theta(t) = t + \theta_0$. This is the expression of the flow in polar coordinates.

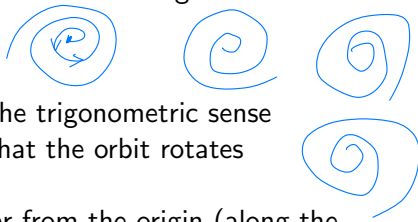
A focus

From the expression of the flow we see that, along an orbit, θ increases (linearly), while ρ increases (exponentially) with respect to the time. Having in mind the geometrical interpretation of the polar coordinates, we see that each orbit is a (logarithmic) spiral that rotates in the trigonometric sense around the origin while departing from it.

In general, we have the following rules.

$\dot{\theta} > 0$ means that the orbit rotates in the trigonometric sense around the origin, while $\dot{\theta} < 0$ means that the orbit rotates clockwise around the origin.

$\dot{\rho} > 0$ means that the state goes further from the origin (along the orbit), while $\dot{\rho} < 0$ means that the state approaches the origin (along the orbit).



The stability of linear planar systems

Theorem

- (i) If $\operatorname{Re}(\lambda_1) < 0$ and $\operatorname{Re}(\lambda_2) < 0$ then the equilibrium point at the origin of (2) is a global attractor (i.e. $A_0 = \mathbb{R}^2$).*
- (ii) If $\operatorname{Re}(\lambda_1) > 0$ and $\operatorname{Re}(\lambda_2) > 0$ then the equilibrium point at the origin of (2) is a global repeller.*
- (iii) Any center is stable.*
- (iv) Any saddle is unstable.*

The *proof* uses that $\varphi(t, \eta) = e^{tA}\eta$ for all $t \in \mathbb{R}$, $\eta \in \mathbb{R}^2$ and is based on the analysis of e^{tA} in each situation.

We will prove only (i).

(i) If $\operatorname{Re}(\lambda_1) < 0$ and $\operatorname{Re}(\lambda_2) < 0$ then the equilibrium point at the origin of (2) is a global attractor.

Proof of (i) in the case that A is diagonalizable over $\mathbb{C}$.

It is sufficient if we find that $\lim_{t \rightarrow \infty} e^{tA}$ is the null matrix.

There exists an invertible matrix P such that $A = PDP^{-1}$, where D is a diagonal matrix with (λ_1, λ_2) on the main diagonal. Then $e^{tA} = Pe^{tD}P^{-1}$ for all $t \in \mathbb{R}$. Also, e^{tD} is a diagonal matrix with $(e^{t\lambda_1}, e^{t\lambda_2})$ on the main diagonal.

The proof is done if we justify that $\lim_{t \rightarrow \infty} e^{t\lambda_k} = 0$ for $k = 1, 2$. This is evident when $\lambda_k \in \mathbb{R}$ since it is strictly negative.

When λ_k is not real we write $\lambda_k = \alpha + i\beta$. Then $e^{t\lambda_k} = e^{t\alpha}e^{it\beta}$.

We have $\lim_{t \rightarrow \infty} e^{t\alpha} = 0$ since $\alpha < 0$ and $e^{it\beta} = (\cos(t\beta) + i\sin(t\beta))$ is bounded. The proof is finished.

The linearization method to study the stability of equilibria of nonlinear planar systems

We consider the nonlinear planar system $\dot{X} = f(X)$, where $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a given C^1 function. Let $\eta^* \in \mathbb{R}^2$ be an equilibrium point. Let $Jf(\eta^*)$ be the Jacobian matrix of f computed in η^* . The linear system

$$\dot{X} = Jf(\eta^*)X$$

is called **the linearization of $\dot{X} = f(X)$ around the equilibrium point η^*** .

The linearization method

Denote by $\lambda_1, \lambda_2 \in \mathbb{C}$ the eigenvalues of $Jf(\eta^*)$.

Definition

We say that the equilibrium point η^* is hyperbolic if $\operatorname{Re}(\lambda_1) \neq 0$ and $\operatorname{Re}(\lambda_2) \neq 0$.

Theorem

Let η^ be a hyperbolic equilibrium point of $\dot{X} = f(X)$.*

If the linear system $\dot{X} = Jf(\eta^)X$ is an attractor/repeller, then the equilibrium point η^* of $\dot{X} = f(X)$ is also an attractor/repeller.*

If the linear system $\dot{X} = Jf(\eta^)X$ is a saddle, then the equilibrium point η^* of $\dot{X} = f(X)$ is unstable.*

Remark

In order to study the stability of the equilibrium points of the second order scalar differential equation

$$\ddot{x} = f(x, \dot{x}),$$

we transform it to a planar system with unknowns x and $y = \dot{x}$. So, the planar system is

$$\dot{x} = y, \quad \dot{y} = f(x, y).$$

$$\dot{y} = \ddot{x} = f(x, \dot{x}) = f(x, y)$$

Stability and First integrals

Theorem

Let η^ be an equilibrium point of $\dot{X} = f(X)$.*

(i) If η^ is an attractor/repeller, then there is no first integral $H: \Omega \rightarrow \mathbb{R}$ such that $\eta^* \in \Omega$.*

(ii) In particular, if the origin is an attractor/repeller of $\dot{X} = AX$, then this linear system does not have a first integral $H: \mathbb{R}^2 \rightarrow \mathbb{R}$ (global first integral).

We prove only (ii).

Stability and First integrals

Proof. (ii) Assume that $\eta^* = 0 \in \mathbb{R}^2$ be an attractor of $\dot{X} = AX$. Then it is a global attractor, i.e.

$$\lim_{t \rightarrow \infty} \varphi(t, \eta) = 0 \quad \text{for all } \eta \in \mathbb{R}^2$$

Assume by contradiction that there exists a first integral H on $\mathbb{R}^2$. Then

$$H(\varphi(t, \eta)) = H(\eta), \quad t \in \mathbb{R}, \quad \eta \in \mathbb{R}^2.$$

Then $\lim_{t \rightarrow \infty} H(\varphi(t, \eta)) = H(\eta)$ for all $\eta \in \mathbb{R}^2$.

Since H is continuous, we get $H(0) = H(\eta)$ for all $\eta \in \mathbb{R}^2$.

This implies that H is constant, which contradicts the definition of the first integral.

A constant function does not distinguish between trajectories and provides no information about the dynamic of the system.

Stability and First integrals

Theorem

Let η^ be an equilibrium point of the planar system $\dot{X} = f(X)$. Assume that the eigenvalues of $Jf(\eta^*)$ are $\lambda_{1,2} \pm i\beta$, with $\beta > 0$ (thus, η^* is not hyperbolic). If $\dot{X} = f(X)$ has a first integral $H \in C^2(\Omega)$ with $\eta^* \in \Omega$, then η^* is a stable equilibrium point.*

Exercise 1 Study the stability of the equilibria of the system using the linearization method:

$$\begin{cases} \dot{x} = x(1-x) = 0 \\ \dot{y} = y(3-y) = 0 \end{cases}$$

Exercise 2: Study the stability of the equilibria of $\ddot{\theta} + \omega^2 \sin \theta = 0$ (the idealized equation of a pendulum)

Important remark: In order to study the stability of the equilibria of a second order nonlinear eq. (scalar) $\ddot{x} = f(x, \dot{x})$, we transform it into a planar system with the unknowns x and $y = \dot{x}$. So the planar system is:

$$\begin{cases} \dot{x} = y \\ \dot{y} = f(x, y) \end{cases}$$

In our case: $x = \theta$ and $y = \dot{\theta}$

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\omega^2 \sin x \end{cases}$$

$(0,0)$, $(1,0)$, $(0,3)$, $(1,3)$

$$Jf(x,y) = \begin{pmatrix} 1-2x & 0 \\ 0 & 3-2y \end{pmatrix}$$

$$Jf(0,0) = \begin{pmatrix} \textcircled{1} & 0 \\ 0 & \textcircled{3} \end{pmatrix} = A$$

$\lambda_2 \in \mathbb{R}$

$(0,0)$ unstable

$$Jf(1,0) = \begin{pmatrix} \textcircled{-1} & 0 \\ 0 & \textcircled{3} \end{pmatrix}$$

$(1,0)$ unstable

$$\ddot{\theta} + \omega^2 \sin \theta = 0$$

$$x = \theta$$

$$y = \dot{x}$$

$$\begin{cases} \dot{x} = y \\ \dot{y} = \ddot{x} = \ddot{\theta} = -\omega^2 \sin \theta \end{cases}$$